imports

TurtleBot
- P IOperations
- i IEvents
- i IYaw
- yaw: real
- scan: Seq(real)

IYaw
- yaw: real

Header

TwistStamped
- header: Header
- twist: Twist

IOperations
- o cmd_vel(msg: TwistStamped)

IEvents
- scan: Seq(real)

IObstacle
- obstacle: Direction

f_x abs(a: real): real

(a>=0=>result==a)/\(a<0=>result==-a)

f_x default(): Header

Twist
- linear: Vector3
- angular: Vector3
- yaw: real

Direction
- left
- right
- front

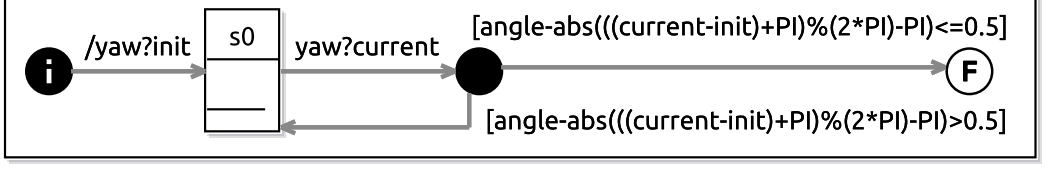
Vector3
- x: real
- y: real
- z: real

IMove
- o move(lvel: real, avel: real)
- o rotate(angle: real)

MoveParams
- π avel: real
- π PI: real

o rotate(angle: real)

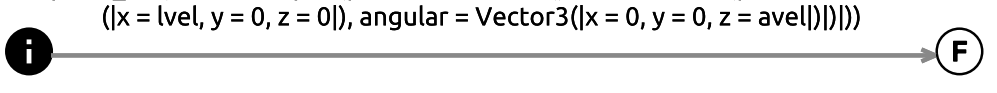
- R MoveParams
- init: real, current: real
- i IYaw



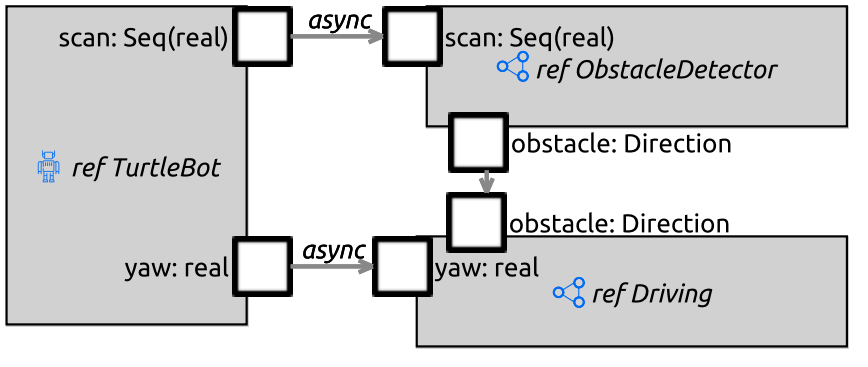
o move(lvel: real, avel: real)

- R IOperations

/cmd_vel(TwistStamped(|header = default(), twist = Twist(|linear = Vector3 (|x = lvel, y = 0, z = 0|), angular = Vector3(|x = 0, y = 0, z = avel|)|)|))

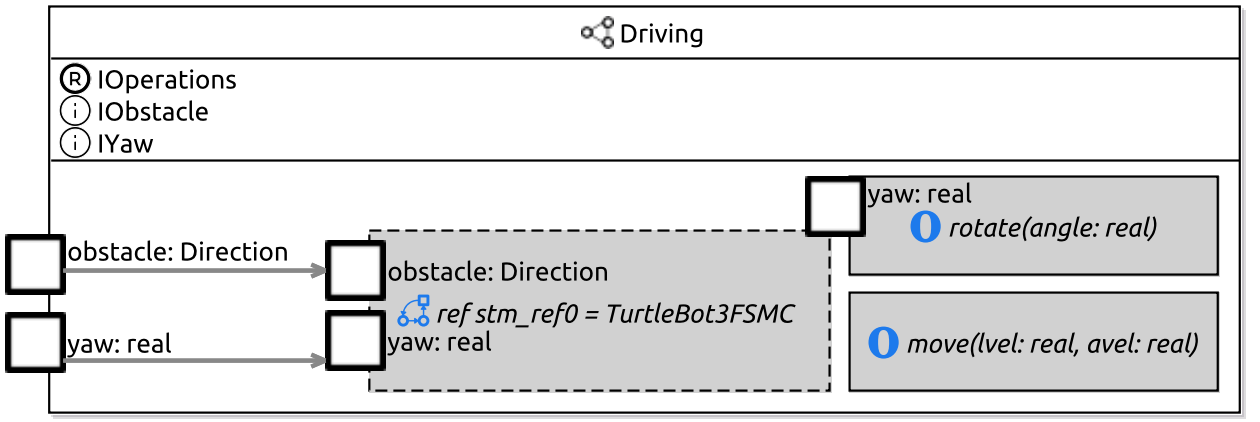


Module



Driving

- R IOperations
- i IObstacle
- i IYaw



obstacle: Direction

yaw: real

TurtleBot3FSMC

- R IOperations
- R IMove
- π lvel: real, avel: real, PI: real
- obs: Direction
- C
- i IObstacle
- i IYaw

